

Built-in (Standard Library) Functions in C – Detailed Notes

1. Input / Output Functions (stdio.h)

- **printf()**: Used to display output on the screen.
- **scanf()**: Used to read input from the keyboard.
- **getchar()**: Reads a single character from standard input.
- **putchar()**: Prints a single character to the screen.
- **puts()**: Prints a string followed by a newline.

2. String Functions (string.h)

- **strlen()**: Returns the length of a string.
- **strcpy()**: Copies one string into another.
- **strcat()**: Appends one string to the end of another.
- **strcmp()**: Compares two strings.
- **strrev()**: Reverses a string (non-standard).

3. Mathematical Functions (math.h)

- **sqrt()**: Returns square root of a number.
- **pow()**: Calculates power of a number.
- **abs()**: Returns absolute value.
- **ceil()**: Rounds number upward.
- **floor()**: Rounds number downward.

4. Character Handling Functions (ctype.h)

- **isalpha()**: Checks if character is an alphabet.
- **isdigit()**: Checks if character is a digit.
- **isalnum()**: Checks if character is alphanumeric.
- **toupper()**: Converts to uppercase.
- **tolower()**: Converts to lowercase.

5. Memory Management Functions (stdlib.h)

- **malloc()**: Allocates memory dynamically.
- **calloc()**: Allocates and initializes memory.
- **realloc()**: Resizes allocated memory.
- **free()**: Releases allocated memory.

6. Utility Functions (stdlib.h)

- **exit()**: Terminates program execution.
- **rand()**: Generates random number.
- **srand()**: Seeds random number generator.
- **atoi()**: Converts string to integer.

7. Time & Date Functions (time.h)

- **time()**: Returns current calendar time.
- **clock()**: Returns processor time.

- **difftime()**: Finds difference between two times.